

Module 1: Frontend Development

Module's	Youtube Link
• Introduction to Web Development	https://youtu.be/HfTXHrWMGVY?si=zSQj4ITv5Jwl_JjB
• HTML5	https://youtu.be/mzPxo7Y6JyA?si=qZPoxvrEkywlU20G
• CSS3	https://youtu.be/OEV8gMkCHXQ?si=E_NsGfsBKcY5mrr
• JavaScript (ES6)	https://youtu.be/NCwa_xi0Uuc?si=sJ1i8E5T3nFKQhHQ
• Responsive Web Design	https://youtu.be/3tLb3i7GB38?si=gF02PJgcLcfN3FIP
• DOM Manipulation	https://youtu.be/NO5kUNxGlu0?si=rJ4k_aqU-7ZYqb_X

1. Introduction to Web Development

Web Development is the process of creating websites and web applications that run on the internet. It involves designing the user interface, writing code, connecting databases, and deploying applications to web servers.

Types of Web Development

- **Frontend Development:** Creates the visual interface users interact with.
- **Backend Development:** Handles server-side logic, databases, and APIs.
- **Full Stack Development:** Combines both frontend and backend development.

How a Website Works

1. User enters a website URL.
2. Browser sends a request to the web server.
3. Server processes the request.
4. Server returns HTML, CSS, JavaScript, and other resources.
5. Browser renders the webpage.

Common Web Technologies

- HTML – Structure
- CSS – Styling
- JavaScript – Interactivity
- Bootstrap – Responsive UI
- APIs – Communication between frontend and backend

2. HTML5

HTML5 (HyperText Markup Language Version 5) is the standard language used to create the structure of web pages.

Features of HTML5

- Semantic elements
- Audio and video support

- Forms with new input types
- Canvas for graphics
- Local storage
- Geolocation API

Basic HTML Structure

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>My Website</title>
```

```
</head>
```

```
<body>
```

```
  <h1>Welcome</h1>
```

```
  <p>Hello World!</p>
```

```
</body>
```

```
</html>
```

Common HTML Elements

- Headings (<h1> to <h6>)
- Paragraph (<p>)
- Image ()
- Link (<a>)
- Table (<table>)
- List (,)
- Form (<form>)
- Button (<button>)
- Div (<div>)
- Section (<section>)

- Header (<header>)
- Footer (<footer>)

Semantic Tags

- <header>
- <nav>
- <main>
- <section>
- <article>
- <aside>
- <footer>

These improve readability and SEO.

3. CSS3

CSS (Cascading Style Sheets) is used to design and style HTML pages.

CSS Types

1. Inline CSS
2. Internal CSS
3. External CSS

Example

```
body{  
    background-color:#f4f4f4;  
    font-family:Arial;  
}
```

```
h1{  
    color:blue;
```

```
}
```

```
button{  
    background:green;  
    color:white;  
}
```

CSS Selectors

- Element Selector
- Class Selector
- ID Selector
- Universal Selector
- Attribute Selector
- Pseudo-class
- Pseudo-element

CSS Box Model

Every HTML element consists of:

- Content
- Padding
- Border
- Margin

CSS Layout Techniques

- Flexbox
- CSS Grid
- Position
- Float
- Display Property

CSS Animations

```
@keyframes move{  
  from{left:0;}  
  to{left:200px;}  
}
```

CSS Transitions

```
button{  
  transition:0.3s;  
}
```

4. JavaScript (ES6)

JavaScript is the programming language used to make websites interactive.

Features of ES6

- let and const
- Arrow Functions
- Template Literals
- Classes
- Modules
- Promises
- Destructuring
- Spread Operator
- Default Parameters

Variables

```
let name="John";  
  
const age=20;
```

Functions

```
function greet(){  
  console.log("Hello");  
}
```

Arrow Function

```
const greet={()=>{  
  console.log("Hello");  
}}
```

Conditional Statement

```
if(age>=18){  
  console.log("Adult");  
}
```

Loop

```
for(let i=1;i<=5;i++){  
  console.log(i);  
}
```

Arrays

```
let fruits=["Apple","Orange","Mango"];
```

Objects

```
let student={  
  name:"John",  
  age:20  
};
```

Events

- Click
- Mouseover
- Submit

- Keypress
- Change

Example:

```
button.addEventListener("click",function(){  
    alert("Button Clicked");  
});
```

5. Responsive Web Design

Responsive Web Design (RWD) ensures that websites automatically adjust to different screen sizes such as desktops, tablets, and smartphones.

Benefits

- Better user experience
- Mobile-friendly
- Improved SEO
- Faster loading
- Single codebase

Responsive Techniques

- Flexible layouts
- Flexible images
- Media Queries
- Bootstrap Grid
- Viewport Meta Tag

Viewport

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

Media Query Example

```
@media(max-width:768px){
```



```
body{  
    background:#ddd;  
}  
}
```

6. DOM Manipulation

DOM stands for **Document Object Model**.

The DOM represents an HTML document as a tree structure, allowing JavaScript to access and modify elements dynamically.

DOM Methods

```
document.getElementById()  
document.querySelector()  
document.querySelectorAll()  
document.createElement()  
document.appendChild()
```

Changing Content

```
document.getElementById("demo").innerHTML="Welcome";
```

Changing CSS

```
document.getElementById("demo").style.color="red";
```

Event Handling

```
document.getElementById("btn").onclick=function(){  
    alert("Clicked");  
}
```

Creating Elements

```
let p=document.createElement("p");  
p.innerHTML="New Paragraph";
```

```
document.body.appendChild(p);
```

7. Bootstrap

Bootstrap is a popular front-end CSS framework used to develop responsive and mobile-first websites quickly.

Advantages

- Responsive Grid System
- Ready-made Components
- Faster Development
- Cross-browser Compatibility
- Customizable
- Mobile First

Bootstrap Grid

Bootstrap divides the screen into **12 columns**.

Example:

```
<div class="row">
```

```
  <div class="col-md-6">
```

```
    Left
```

```
  </div>
```

```
  <div class="col-md-6">
```

```
    Right
```

```
  </div>
```

```
</div>
```

Common Bootstrap Components

- Navbar

- Buttons
- Cards
- Forms
- Alerts
- Modal
- Carousel
- Accordion
- Tables
- Pagination
- Dropdown
- Badge
- Spinner

Bootstrap Button Example

```
<button class="btn btn-primary">  
  Submit  
</button>
```

Bootstrap Card Example

```
<div class="card">  
  <div class="card-body">  
    Welcome to Full Stack Development  
  </div>  
</div>
```

Bootstrap Form Example

```
<input type="text" class="form-control" placeholder="Enter Name">
```